StudySapiens Complete Chapter Notes

Chapter 4 — Biomolecules

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 4.1 to 4.9. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 4 carries 2 MCQs, 3 short questions in Q.4, and part (a) of a long question in Q.9. Shorts are 2 marks; longs are 8 marks.

TOPIC 4.1 — BIOLOGICAL MOLECULES

Textbook topic 4.1 · printed pages 91–93

Q.1 Explain the importance of organic molecules and macromolecules in living organisms.

Board long question · 8 marks

Life on Earth evolved in water, and all life still depends on water. The molecules synthesized by cells and containing carbon are known as **organic molecules**. Carbohydrates, proteins, lipids and nucleic acids are important organic molecules and make up about **93% of the dry mass** of living organisms. The remaining 7% comprises small organic molecules such as vitamins and inorganic molecules such as carbon dioxide, acids, bases and salts.

Most organic molecules are large and are called **macromolecules**. Many are polymers, consisting of many identical molecular units called **monomers**. Carbohydrates, proteins and nucleic acids are polymers of sugars, amino acids and nucleotides respectively. Organic molecules have a carbon-based core with functional groups such as –OH, –CO, –COOH and –NH2. Most biochemical reactions involve transfer of functional groups or breaking of carbon-carbon bonds.

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Important Conceptual Short Questions — Topic 4.1

S1. Why is carbon considered the basic element of organic molecules?

Carbon is tetravalent and can bond with many elements such as H, O, N, P and S, forming a carbon-based core with diverse functional groups.

S2. How do monomers and polymers relate to biological macromolecules?

Polymers are large molecules made of many repeating monomers. Carbohydrates, proteins and nucleic acids are polymers of sugars, amino acids and nucleotides respectively.

S3. Why do functional groups determine the behaviour of organic molecules?

Most biochemical reactions involve transfer of functional groups or breaking of carbon-carbon bonds, so functional groups determine reactivity and properties.

TOPIC 4.2 — TYPES OF BONDS IN BIOLOGY

Textbook topic 4.2 · printed pages 93–94

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Q.2 Describe the types of bonds and interactions important in biological molecules.

Board long question · 8 marks

Bond / interaction	Formation	Biological importance
Covalent bond	Two atoms share electrons	Stability of organic molecules such as proteins and nucleic acids
Ionic bond	One atom donates an electron and another accepts it; electrostatic attraction between ions	Strong in solid state; common in inorganic molecules like NaCl
Hydrogen bond	Weak attraction between hydrogen and an electronegative atom such as O or N	Maintains structure of proteins and nucleic acids; important in DNA replication
Hydrophobic interaction	Nonpolar molecules cluster together in aqueous environments	Protein folding and formation of lipid bilayers
Hydrophilic interaction	Polar molecules interact with water	Dissolution of polar and ionic compounds; nutrient transport and reactions

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Important Conceptual Short Questions — Topic 4.2

S1. Why are covalent bonds important in biological molecules?

They provide stability by sharing electrons between atoms, forming the backbone of proteins, nucleic acids and other organic molecules.

S2. How do hydrogen bonds contribute to DNA structure?

They form weak attractions between complementary bases, helping maintain the double helix and allowing strands to separate during replication.

S3. Why do hydrophobic interactions drive protein folding and membrane formation?

Nonpolar molecules cluster together to minimize contact with water, helping proteins fold and lipids form bilayers in membranes.

S4. Differentiate between hydrophilic and hydrophobic interactions.

Hydrophilic interactions occur between polar molecules and water, aiding dissolution. Hydrophobic interactions occur between nonpolar molecules that cluster together to avoid water.

TOPIC 4.3 — CONDENSATION (SYNTHESIS) AND HYDROLYSIS

Textbook topic 4.3 · printed pages 94–95

Q.3 Differentiate between condensation and hydrolysis. How do they build and break macromolecules?

Board long question · 5 + 3 marks

Feature	Condensation (dehydration synthesis)	Hydrolysis
Purpose	Builds polymers from monomers	Breaks polymers into monomers
Water	Removed: –OH from one monomer and –H from another	Added: –OH joins one monomer and –H joins the other
Energy	Requires energy input	Releases smaller molecules for use
Example	Two glucose molecules form maltose	Maltose breaks into two glucose molecules

Cells assemble proteins, nucleic acids, carbohydrates and lipids by condensation. They also keep breaking polymers by hydrolysis, which is essential in digestion and cellular respiration.

Important Conceptual Short Questions — Topic 4.3

S1. Why is condensation also called dehydration synthesis?

Because an –OH group and an –H atom are removed from monomers, resulting in the removal of a water molecule during bond formation.

S2. How does hydrolysis help in digestion?

It breaks macromolecules into smaller fragments by adding water, producing absorbable monomers such as glucose and amino acids.

S3. Why do cells need to supply energy for condensation reactions?

Energy is required to form bonds between monomers and remove water during synthesis of macromolecules.

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TOPIC 4.4 — IMPORTANCE OF WATER

Textbook topic 4.4 · printed pages 95–99

Q.4 Explain the properties of water that make it essential for life.

Board long question · 8 marks

- **Solvent properties:** Water is polar and forms hydrogen bonds. It dissolves charged or polar substances (hydrophilic) such as salts, sugars and amino acids, providing a medium for metabolism. Nonpolar substances such as lipids are hydrophobic and are excluded from water.
- **High specific heat capacity:** Water absorbs or releases much heat with minimum change in its own temperature (4.184 J/g°C). It acts as a temperature stabilizer for organisms and their environment.
- **High heat of vaporization:** A large amount of heat (574 kcal/kg) is required to change liquid water into vapour. Evaporation provides cooling in transpiration and perspiration.
- **Cohesion:** Hydrogen bonds make water molecules stick together. Cohesion helps transport of water and nutrients from roots to leaves against gravity.
- **Surface tension:** Hydrogen bonds give water high surface tension, allowing insects such as water striders to walk on its surface.
- **Hydrophobic exclusion:** Nonpolar molecules are pushed out of water and associate with one another, helping maintain cell membranes.

Important Conceptual Short Questions — Topic 4.4

S1. Why is water called a universal solvent for biological systems?

Its polarity and hydrogen bonding allow it to dissolve many charged and polar substances, providing a medium for metabolic reactions.

S2. How does high specific heat capacity of water help organisms?

It allows water to absorb or release large amounts of heat with little temperature change, stabilizing internal and external environments.

S3. Why does evaporation of sweat cool the body?

Water has a high heat of vaporization, so each gram of evaporating water removes a large amount of heat from the body surface.

S4. How does cohesion help in the ascent of sap?

Hydrogen bonds make water molecules stick together, so the pulling force from leaf evaporation is relayed through xylem vessels down to the roots.

S5. Why are lipids insoluble in water?

Lipids are nonpolar and cannot form hydrogen bonds with water. Water molecules exclude them, causing hydrophobic association.

S6. How does hydrophobic exclusion help in membrane formation?

Nonpolar lipid molecules cluster together away from water, forming stable lipid bilayers that make up cell membranes.

TOPIC 4.5 — CARBOHYDRATES

Textbook topic 4.5 · printed pages 99–106

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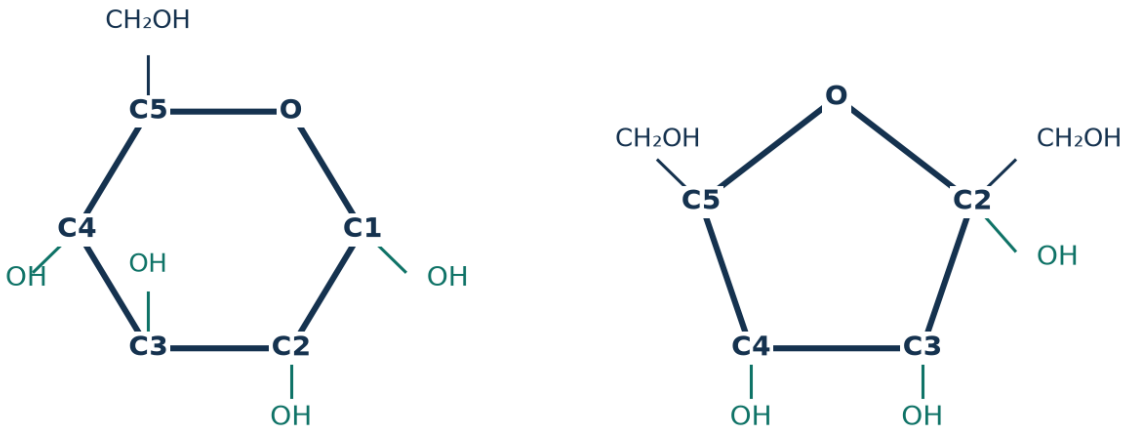
Q.5 Classify carbohydrates and describe monosaccharides, disaccharides and polysaccharides.

Board long question · 8 marks

Classification

- **Monosaccharides:** Simple sugars such as glucose, fructose, galactose and ribose. They may exist as straight chains or ring structures.
- **Disaccharides:** Formed by two monosaccharides joined by a glycosidic linkage. Examples: maltose (glucose + glucose), lactose (glucose + galactose), sucrose (glucose + fructose).
- **Polysaccharides:** Long chains of monosaccharides. Examples: starch, glycogen, cellulose and chitin.

α-D-glucose (six-membered ring) Fructose (five-membered ring)



Ring structures of glucose and fructose.

Important polysaccharides

Polysaccharide	Structure	Function
Starch	Amylose (unbranched glucose chain) and amylopectin (branched glucose chain)	Storage carbohydrate in plants
Glycogen	Highly branched glucose chain, similar to amylopectin but more branched	Storage carbohydrate in animals
Cellulose	Unbranched glucose chains with beta linkages	Structural component of plant cell walls
Chitin	Polymer of nitrogen-containing sugar	Structural component of fungal walls and arthropod exoskeletons

Important Conceptual Short Questions — Topic 4.5

S1. Why do glucose and fructose have the same formula but different properties?

They are structural isomers. Glucose is an aldose and fructose is a ketose, so their functional groups and ring structures differ.

S2. How does a glycosidic linkage form between monosaccharides?

It forms by a condensation reaction in which a water molecule is removed and an oxygen bridge joins two sugar molecules.

S3. Why is glycogen more branched than starch?

Glycogen is the animal storage carbohydrate and its highly branched structure allows rapid release of glucose when energy is needed.

S4. Differentiate between amylose and amylopectin.

Amylose is an unbranched chain of glucose, while amylopectin is a branched chain of glucose. Both are components of starch.

S5. Why can humans digest starch but not cellulose?

Humans have enzymes that break alpha glycosidic linkages in starch but lack enzymes to break the beta linkages in cellulose.

S6. How does chitin provide structural support?

It is a tough nitrogen-containing polysaccharide that forms fungal cell walls and arthropod exoskeletons, giving strength and protection.

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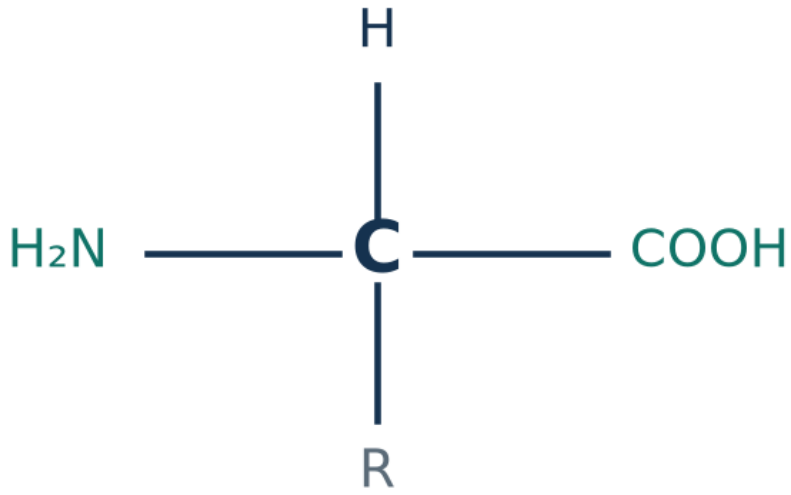
TOPIC 4.6 — PROTEINS

Textbook topic 4.6 · printed pages 106–113

Q.6 Describe the structure of amino acids and the formation of proteins.

Board long question · 8 marks

Proteins are polymers of **amino acids**. Each amino acid has a central carbon atom bonded to an amino group (–NH2), a carboxyl group (–COOH), a hydrogen atom and a variable **R group**. The R group determines the properties of the amino acid.



General amino acid (α-carbon with –NH₂, –COOH, –H and R group)

General structure of an amino acid.

Amino acids join by **peptide bonds** formed through condensation between the carboxyl group of one amino acid and the amino group of another. A chain of amino acids is called a **polypeptide**.

Structural levels of proteins

Level	Description
Primary structure	Specific number and sequence of amino acids in a polypeptide chain
Secondary structure	Local folding into alpha-helix or beta-pleated sheet, stabilized by hydrogen bonds
Tertiary structure	Overall three-dimensional folding of a polypeptide, stabilized by various bonds and interactions
Quaternary structure	Association of two or more polypeptide chains, e.g., haemoglobin

Q.7 Classify proteins and describe their roles in life.

Board long question · 8 marks

Basis	Types	Examples / roles
Structure	Fibrous (primary/secondary) and globular (tertiary/quaternary)	Fibrous: structural; globular: functional
Solubility	Insoluble fibrous; soluble globular	Globular proteins soluble in salt, acid or base
Role	Structural and functional	Collagen, actin; enzymes, hormones, antibodies

Proteins carry out virtually all cellular functions. They act as **enzymes**, **structural components**, **transport molecules**, **hormones**, **antibodies** and **contractile proteins**. A change in amino acid sequence can alter protein function, as in sickle cell haemoglobin.

Important Conceptual Short Questions — Topic 4.6

S1. Why does the R group determine the properties of an amino acid?

The R group varies among amino acids and determines whether they are polar, nonpolar, acidic or basic, influencing protein structure and function.

S2. How does a peptide bond form between amino acids?

It forms by condensation between the carboxyl group of one amino acid and the amino group of another, releasing a water molecule.

S3. Why is the primary structure of a protein critical for its function?

The sequence of amino acids determines folding and final three-dimensional shape. A change in sequence can alter function, as in sickle cell haemoglobin.

S4. Differentiate between secondary and tertiary structure of proteins.

Secondary structure is local folding into alpha-helix or beta-pleated sheet stabilized by hydrogen bonds. Tertiary structure is the overall three-dimensional folding of the whole polypeptide.

S5. Why do globular proteins generally function as enzymes and hormones?

Their compact, soluble three-dimensional shapes allow them to interact specifically with other molecules, making them suitable for catalysis and signalling.

S6. How does denaturation affect protein function?

Denaturation disrupts the three-dimensional structure of a protein, causing loss of its specific function.

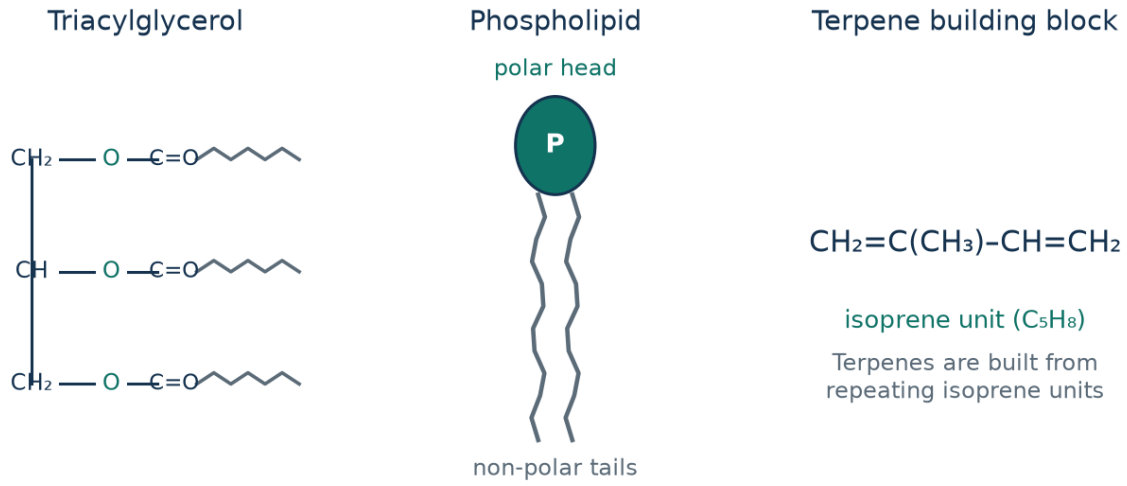
TOPIC 4.7 — LIPIDS

Textbook topic 4.7 · printed pages 113–118

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Q.8 Describe the types of lipids and their roles in life.

Board long question · 8 marks



Representative lipid structures.

- **Acylglycerols (fats and oils):** Formed from glycerol and fatty acids. Triacylglycerols store energy. Saturated fats are solid at room temperature; unsaturated oils are liquid.
- **Phospholipids:** Contain glycerol, two fatty acids and a phosphate group. They are amphipathic and form the lipid bilayer of membranes.
- **Terpenes:** Built from isoprene units. Include pigments, vitamins and steroids.
- **Steroids:** Have a four-ring structure. Examples include cholesterol and sex hormones.
- **Prostaglandins:** Lipid compounds involved in inflammation and other local signalling processes. Aspirin inhibits their synthesis.

Lipids function in **energy storage**, **membrane structure**, **insulation**, **protection** and **signalling**.

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Important Conceptual Short Questions — Topic 4.7

S1. Why are phospholipids able to form bilayers in water?

They are amphipathic: hydrophilic phosphate heads face water while hydrophobic fatty acid tails face inward, forming a stable bilayer.

S2. Differentiate between saturated and unsaturated fats.

Saturated fats have no double bonds in fatty acid chains and are usually solid at room temperature. Unsaturated fats have one or more double bonds and are usually liquid oils.

S3. Why are triacylglycerols efficient energy storage molecules?

They contain many carbon-hydrogen bonds that release large amounts of energy on oxidation, and they are stored in anhydrous form.

S4. How do steroids differ structurally from other lipids?

Steroids have a characteristic four-ring carbon structure, unlike the fatty acid chains of acylglycerols and phospholipids.

S5. Why does aspirin reduce inflammation?

Aspirin inhibits the synthesis of prostaglandins, which are lipid compounds involved in inflammation and pain signalling.

TOPIC 4.8 — NUCLEIC ACIDS

Textbook topic 4.8 · printed pages 118–123

Q.9 Describe the structure of nucleic acids and differentiate between DNA and RNA.

Board long question · 8 marks

Nucleic acids are polymers of **nucleotides**. Each nucleotide consists of a **pentose sugar**, a **phosphate group** and a **nitrogenous base**. Nucleotides join by **phosphodiester bonds** to form polynucleotide chains.

Feature	DNA	RNA
Sugar	Deoxyribose	Ribose
Bases	Adenine, guanine, cytosine, thymine	Adenine, guanine, cytosine, uracil
Strands	Usually double-stranded	Usually single-stranded
Function	Stores genetic information	Helps in protein synthesis

Q.10 Explain the double helical structure of DNA as proposed by Watson and Crick.

Board long question · 8 marks

Watson and Crick proposed the double helix model of DNA in 1953, using data from Rosalind Franklin and Maurice Wilkins. DNA consists of **two antiparallel polynucleotide chains** coiled around a common axis to form a right-handed double helix.

- The sugar-phosphate backbones lie on the outside, while nitrogenous bases face inward.
- Bases pair specifically: **adenine with thymine** (two hydrogen bonds) and **guanine with cytosine** (three hydrogen bonds).
- The two strands are complementary and antiparallel, running in opposite directions.
- The diameter of the helix is about 2 nm, and one complete turn contains about 10 base pairs.

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This structure explains how DNA stores genetic information and how it can be replicated by separation of strands and complementary base pairing.

Important Conceptual Short Questions — Topic 4.8

S1. Why is DNA more stable than RNA?

DNA contains deoxyribose, which lacks the reactive 2′ hydroxyl group present in ribose, and its double-stranded structure provides additional stability.

S2. How do phosphodiester bonds form the backbone of nucleic acids?

They join the phosphate group of one nucleotide to the sugar of the next, creating a repeating sugar-phosphate backbone.

S3. Why do adenine pair with thymine and guanine with cytosine?

Their shapes and hydrogen-bonding patterns allow specific complementary base pairing: A–T forms two hydrogen bonds and G–C forms three.

S4. How does the antiparallel nature of DNA strands affect replication?

Each strand runs in opposite direction, so new strands are synthesized in opposite directions, allowing each strand to serve as a template.

S5. Why is the double helix structure important for genetic information storage?

The sequence of bases stores information, while complementary base pairing allows accurate copying and transmission of genetic information.

S6. Differentiate between the roles of DNA and RNA.

DNA stores genetic information. RNA helps in protein synthesis by carrying instructions and assisting in translation.

TOPIC 4.9 — CONJUGATED MOLECULES

Textbook topic 4.9 · printed pages 124

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Q.11 Describe conjugated molecules and give their biological importance.

Board long question · 8 marks

Conjugated molecule	Composition	Importance
Glycoproteins	Protein + carbohydrate polymer (covalent linkage)	Membrane structure, blood serum, cellular secretions, cartilage, eyes, skin
Glycolipids	Lipid + carbohydrate (covalent linkage)	Integral structural component of membranes
Lipoproteins	Lipid + protein (hydrophobic interactions)	Structural framework of plasma membranes; transport of lipids in blood
Nucleoproteins	Nucleic acid + protein	Chromosomes and ribosomes

Important Conceptual Short Questions — Topic 4.9

S1. Why are glycoproteins important on cell surfaces?

They act as identification marks and receptors, helping in cell recognition and signalling.

S2. How do lipoproteins help in lipid transport?

Lipids are hydrophobic and cannot travel freely in blood. Lipoproteins package lipids with proteins for transport in the bloodstream.

S3. Differentiate between glycolipids and glycoproteins.

Glycolipids are lipid-carbohydrate conjugates, while glycoproteins are protein-carbohydrate conjugates. Both are important in membrane structure and cell recognition.

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